14/6 (iii) When Subnet Are Not Connected Through Route

Aim:

To analyze Packet transmission between different subnet when no router is avalaible.

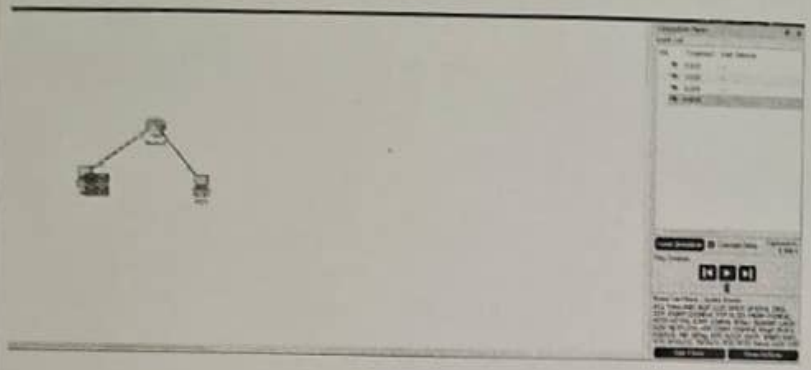
Steps:-

1. Remove the router from the network topology
2. Open simulation mode
3. Select device belonging to different subnet.
4. Generate Packet using the simple PDU tools
5. Observe the Packet transmition
6. Perform a ping the between device located a ping in different subnet
7. Analyze the Packet flow
8. Observe the simulation event
9. Identify the Point at which Packet transmition fail
10. Compare the result with the network that is a router.
11. Record the observation
12. Analyze the importance of a router in inter-network communication

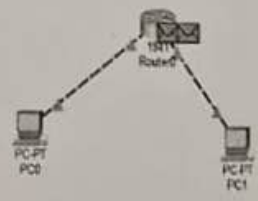
Result:

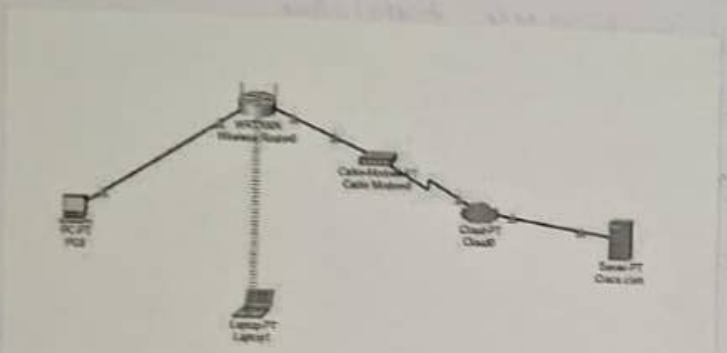
This communication between device located in different subnet fild when router was not used. The Packet could not reached the destination. This emp demonstrated the importance of router in enabling communication between different subnetworks.

Expt 10a










```
Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet Connection (default gate)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.9689.2B1A
   Link-local IPv6 Address.........: FE80::201:96FF:FE89:2B1A
   IP Address......................: 192.168.0.101
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.0.1
   DNS Servers.....................: 208.67.220.220
   DHCP Servers....................: 192.168.0.1
   DHCPv6 Client DUID..............: 00-01-00-01-7D-3B-23-3C-00-01-24-22-29-1B
```

```
C:\>ping Cisco.com

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=10ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms
```

Expt - 10a
17/9/24

Internet Working With Routers In Cisco Packet Tracer

Aim:

To design and configure a simple internetwork using a router and to observe connectivity between two PCs

Steps:

1. Create the network topology
   → Open cisco packet tracer and create a new Project
2. Place one router and two PCs.
3. Connect PC0 and PC1 to the router using copper straight through cables
4. Configure the router:
   - Select the router and open CLI
   - Press Enter
   - Type enable to enter privileged mode
   - Configure the router interface as follows.

Router > enable
router (Config) # interface fastethernet 0/0.
router (config-if) # ip address 192.168.10.1 255.255.255.0
router (config-if) # no shutdown.
router (config) # interface fastethernet 0/1
router (config-if) # ip address 192.168.20.1 255.255.255.0
router (config-if) # no shutdown.

5. Configure the PC's:-
   - Open each PC → desktop → IP configuration.
   - Assign the following address

4a) SIMULATE PACKET TRANSMISSION ACROSS THE
12/6/24 SUBNET AND OBSERVE THE RESULT

i) When subnet are connected through a router

Aim: To analyze Packet transmission b/w different subnet when a router is used for communication

Steps:

1. Open simulator mode in Cisco Packet tracer
2. Select a source device from one subnet from.
3. Select a destination device from another subnet.
4. Generate a Packet using the simple PDU tool
5. Observe the movement of the Packet through the network
6. Analyze the how router sends the Packet
7. Observe the forwarding process performed by the router
8. Verify the Packet transmission between different subnet
9. Use the ping command to test connectivity.
10. Observe the successful delivery of Packets
11. Record the simulation result
12. Analyze the routing process.

Result:

Thus, Packet were successfully transmitted between different subnet through the router. The router received the Packet from the source subnet, processed, them and forwarded them it the destination subnet.